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Introduction

Software version 2021R1.1 continues improvements to the general functionality, speed and usability of **RedHawk.**

A. General Enhancements

Setup Design

- Enhanced RedHawk to support the FOREIGN statements in LEF files. Added the GSR keyword "IGNORE_LEF_FOREIGN_OFFSET" that enables the feature when set to 0.

  Syntax:

  ```
  IGNORE_LEF_FOREIGN_OFFSET [0|1]        (Default:1)
  ```

Power Calculation

- Enhanced the GSR keyword, POWER_LEAKAGE_SCALING_FILE, to support domain-based leakage power scaling.

  Syntax:

  1) `<cell_name> <leakage_scaling_factor>`

     Applied to both 1p1g and multi-rail cell to scale leakage power for all p/g pins.

  2) `<cell_name> <leakage_scaling_factor> <power_pin_name>`

     Applied to multi-rail cell to scale leakage power for the specified power pin.

  3) `<cell_name> <leakage_scaling_factor> <power_domain_name> N`

     Applied to 1p1g cell to scale leakage power for the specified power domain.

     'N' is required for power domain name.

  4) `<cell_name> <leakage_scaling_factor> <power_pin_name> <power_domain_name> N`

     Applied to multi-rail cells to scale leakage power for the specified power domain.

     This must be defined by pin. 'N' is required for power domain name.

  * Wildcards(*) are supported only for cell names.

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* Entries without wild card specification have higher priority over entries with wild card specification.

Examples:

```
1)  Cell1 0.4
2) * 0.8 VDDA_pin
   * 0.6 VDDB_pin
3) * 0.8 VDD_domain N
4) Cell2 0.4 VDDA_pin VDDA_domain N
   Cell2 0.6 VDDB_pin VDDB_domain N
```

Extraction

- Enhanced Tech file keyword 'RPV_VS_AREA' to support extrapolation for VIA areas. ircx2tech will inserts a new keyword ‘USE_R_EXTRAPOLATION 1 ’when extrapolation is needed.

Example:

```
        RPV_VS_AREA {
         {<min-area_1> <resistance_1>
         ...
          <max-area_n> <resistance_n>
     ...     }
     }
     USE_R_EXTRAPOLATION 1
```

ESD

- ESD rule syntax has been enhanced to support multiple zapping sources at user specified locations in one ESD-CD test. User can specify the source and zap_from/to in sections enclosed by new ESD rule keywords, BEGIN_ZAP and END_ZAP.

Syntax:

```
BEGIN_ESD_RULE
 NAME <name>
 TYPE CD

 BEGIN_ZAP
  ZAP_CURRENT <current_in_Amps> # # or use ZAP_VOLTAGE/ZAP_R
# ZAP_VOLTAGE <voltage_in_Volts>
# ZAP_R <resistance_Ohms>
# zap from/to definition; e.g.:
  BUMP_PAIR <bump1> <bump2>
 END_ZAP

 BEGIN_ZAP
  ...
 END_ZAP
 ...
END_ESD_RULE
```

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1) Zap source (current, or both voltage and resistance) is required within the BEGIN_ZAP and END_ZAP section.

2) The following set of existing ESD rule keywords are supported for 'zap from/to definition' in BEGIN_ZAP/END_ZAP (PIN (or pin) for CPA package pin (names)):

```
BUMP_PAIR <bump1> <bump2>
NET_PAIR <net1> <net2>
PIN_PAIR <pin1> <pin2>
ZAP_BUMP_BUMP <bump1> <bump2>
ZAP_BUMP_CLAMP <bump1> <cinst> <locID>
ZAP_BUMP_INST <bump> <inst> <net>
ZAP_CLAMP_BUMP <cinst> <locID> <bump>
ZAP_CLAMP_CLAMP <cinst1> <locID1> <cinst2> <locID2>
ZAP_CLAMP_INST <cinst> <locID> <inst> <net>
ZAP_CLAMP_PIN <cinst <locID> <pin>
ZAP_INST_BUMP <inst> <net> <bump>
ZAP_INST_CLAMP <inst> <net> <cinst> <locID>
ZAP_INST_INST <inst1> <net1> <inst2> <net2>
ZAP_INST_PIN <inst> <net> <pin>
ZAP_PIN_CLAMP <pin> <cinst> <locID>
ZAP_PIN_INST <pin> <inst> <net>
ZAP_PIN_PIN <pin1> <pin2>
ZAP_FROM_BUMP <bump>
ZAP_FROM_CLAMP <cinst> <locID>
ZAP_FROM_INST <inst> <net>
ZAP_FROM_PIN <pin>
ZAP_FROM <name> <x> <y> [<layer> [<net>]]
ZAP_TO_BUMP <bump>
ZAP_TO_CLAMP <cinst> <locID>
ZAP_TO_INST <inst> <net>
ZAP_TO_PIN <pin>
ZAP_TO <name> <x> <y> [<layer> [<net>]]
```

- Enhanced PathFinder to support the rule, `TIE_CLAMP`, with B2B CD check.
- Modified the default maximum current and voltage values for CECM spice model to 10000 and added ESD clamp file options for user to control it.

  For Transmission Line Pulse (TLP) IV, if user wants to use the max TLP values in CECM model, use the keyword:

  ```
  CECM_TLP_BOUND 1
  ```

  To control the range for any type of IV, use the following keywords:

  ```
  CECM_I_BOUND <I1> <I2>
  CECM_V_BOUND <V1> <V2>
  ```

  I1/V1 is the value for 'forward' direction and the I2/V2 for 'reverse' direction. These bounds only have effect to the 1st and the last pwl segment, I has higher priority over V.

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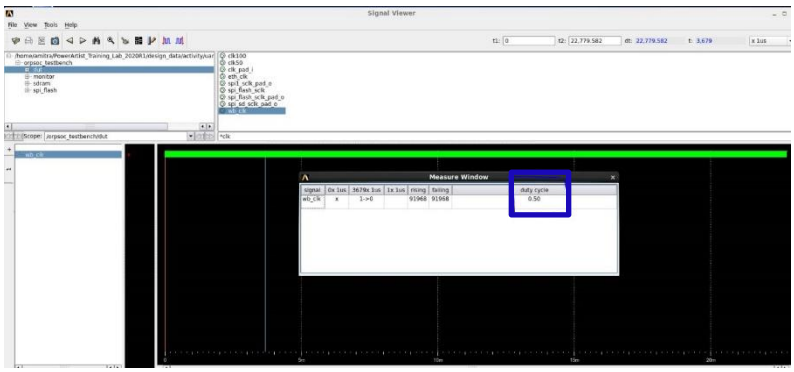
- Enhanced Pathfinder to allow `ESD_CLAMP_PIN_NODE_DISTANCE` to control node creation when clamp nodes are shorted using the keyword `ESD_SHORT_CLAMP_PIN`. This provides a more uniform channel current distribution for clamp devices.
- Improved the process speed of PathFinder-S B2C resistance report generation in DMP-DB flow .
- PathFinder-S DMP flow has been enhanced to support configuration keyword 'GRID_TYPE RSH' which is for the RSH grid. The behavior with 'GRID_TYPE RSH' works the same way as with 'GRID_TYPE SSH'.
- For PathFinder- DMP, in the config file specified with Tcl command `'perform esdcheck -rule <rulefile> -jobFile <jobfile>'`,for GRID_TYPE SSH and RSH, it no longer requires the first machine specified in `CANDIDATE_HOST_LIST` to be the host machine for the RedHawk master process.

GUI

- Enhanced SignalViewer to measure the duty cycle of a signal or an expression. By default SignalViewer will now show the duty cycle with respect to the whole duration of the simulation window but user can choose a specific window using cursors and accordingly duty cycle nd toggle count will dynamically change as part of the ‘Measure’ window.



- Added a new GSR keyword `DISPLAY_HIGHEST_PIN_CURRENT` to display the highest pin (power or ground pin) current of an instance in the GUI when set to 1.

  Syntax:

```
DISPLAY_HIGHEST_PIN_CURRENT [1 |0]    (Default:0)
```

DMP

- Modified the minimum value of DMP configuration keyword, RED_GRID_WIDTH_IN_UM, from 100 µm to 10µm. This will fix the issue where RedHawk failed to honor partitions with small widths.

CTA

Support New Syntax in Thermal Tech file

Support the section of "SOIC_BONDING" in Thermal Tech file

Support Ruler

In this release, user can use Ruler in CTA layout view and 3D postprocessing dialog box. To user Ruler, for example, user can right-click and select Ruler in the pop out menu. Place the mouse over the starting point, when the mouse becomes a diamond, click the left button, move the mouse, at the

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second point, when the mouse becomes a diamond, click the left button again, then user can get the distance. (Figure 1)

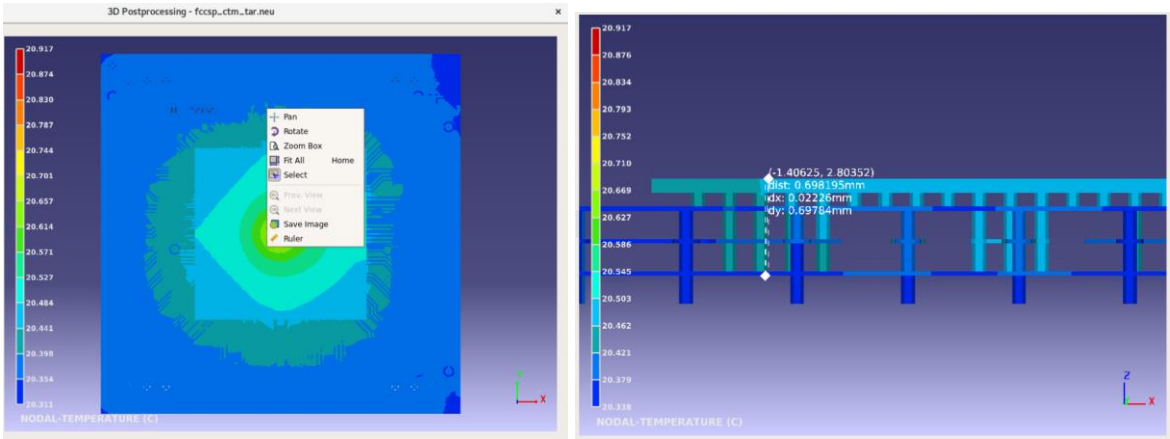

Figure 1 Ruler

Support Thermal Tech File in GUI

Enhanced CTA to support thermal tech file importing on CTA **Setup -> Simulation tab -> Thermal Tech File** button. (Figure 2)

In **TIM and Molding** setting, select “Use TIM” or “Use Molding” in the check-box, the TIM information or the Molding information in the project will be replaced, CTA will use the TIM information or molding information defined in the tech file.

In **Die Mapping**, select the TF die or TF bump, CTA will use the die information or bump information from the tech file.

In **Component Material Mapping** setting, only when the component type is other, it can be displayed.

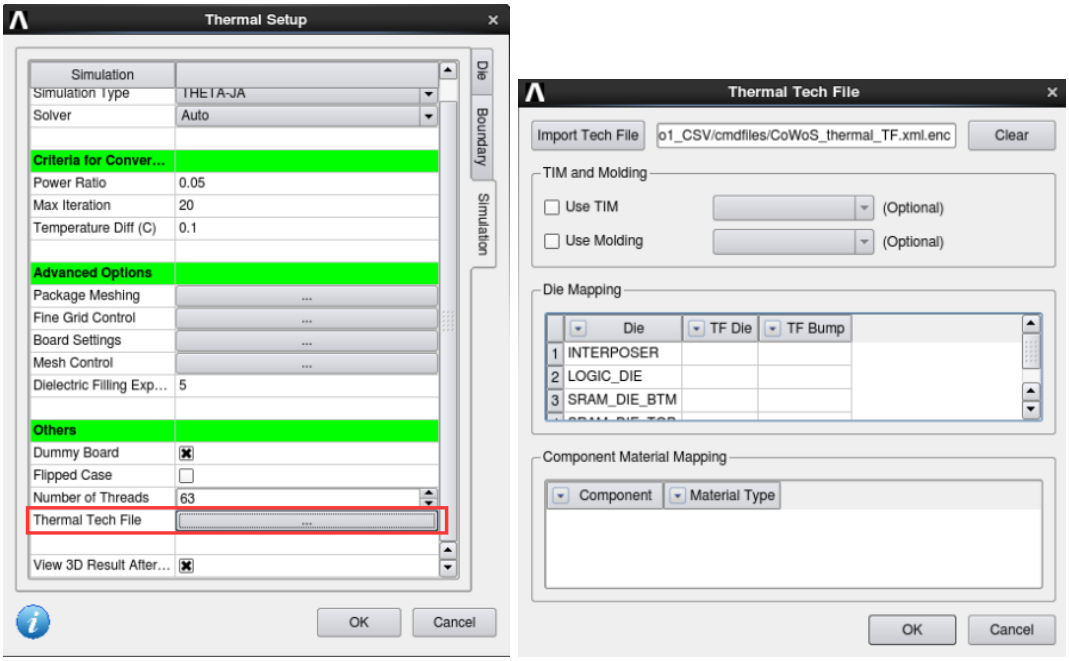

Figure 2 Import Thermal Tech File

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Dump MHS File with Temperature Information

Enhanced CTA to dump MHS file with temperature per tile for chip. This file is in project/ /thermal/adsRpt/CTA/, and the file format is `ThermalProfile_<die_name>:<material>.mhs`

The MHS format is as shown:

```
#<BBOX_id> <llx> <lly> <ux> <uy> <Temp>
2008 9.275 -409 29.275 -399 28.3

2009 19.275 -399 29.275 -389 30.4
```

New TCL Command

Set TSV Material

```
setup cpa thermal -die <name> -tsv <file> ?-model [line | solid]? ?-
diameter <value(mm)>? ?-material <name>?
```

-die: specifies die component name.

-type [conductor | dielectric]: specifies the material type. For conductor, only conductivity is needed.For dielectric materials, relative permittivity, relative permeability, loss tangent and conductivity need to be specified.

-model <line | solid>: specifies the line model or solid model

-diameter: specified tsv diameter.

-material: specified tsv material

Examples:

1. Add a new material “my_copper”.
   ```
   setup cpa material -die U1 -tsv /home/user/test.tsv -model line -
   diameter 0.032 -material my_copper
   ```

Selfheat

- Selfheat thermal calculation includes enhancements as shown below:
  - Support to consider TSV joule heat in thermal aware DC EM flow.
  - Support new SH file parser for SMIC new spec.
  - Support new DT_W2W calculation in flow4 for SMIC
  - Support new DT_FEOL_Coupled calculation in flow4 for SMIC
  - Support new DT_OD calculation in flow4 for SMIC
- Support new DT_WS calculation in flow4 for SMIC.
- Enhanced Selfheat GUI to display the deltaT components of vias post-thermal click/query.

Tcl

- Enhanced Tcl command “plot charge” to use Signal Viewer (sv) as default waveform viewer.

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Other

- **aplmerge.** Added an option -icheck to convert the error message ACK-096 to a warning message during standalone `aplmerge` run.

  Syntax:

  ```
  aplmerge {multiple}.current -icheck
  ```

- **sim2iprof.** Added an option -icheck to convert error message ACK-096 to a warning message.

  Syntax:

  ```
  sim2iprof {multiple}.conf -icheck
  ```

- apldi2 has been enhanced to handle certain types of complex latch cells.
- Added an option `-enc` to `rhtech` utility to automatically insert partial encryption flags `#ENCRYPT_START/#ENCRYPT_END` to the non-EM part.

  Syntax:

  rhtech -i <foundry ITF file > **-enc** -pe <EM tech file> -f <foundry name> -n <process node>

- Enhanced apldi utility to capture spice run time for Spectre based APL characterization.
- ACE has been enhanced to handle spice model files that has sub-circuits defined within a sub-circuit, and the secondary level sub-circuits has the same name as the primary but with a different number of ports.
- Improved EM flow to write out lesser CPU/memory usage information into the debugging file .apache/.nx.log which will reduce the diskspace usage

B. Issues Resolved

Setup Design

- Fixed a bug where metal geometries were not correctly retained in the defined region of the GSR keyword "DESIGN_IMPORT_REGION".
- Fixed a bug where, port name 'CLOCK' with 'L' (load) entry in STA_FILE (or in signal_load file from ENABLE_ATE) was being incorrectly reported for syntax error.

Power Calculation

- Fixed a bug in state propagation that generated false LIB-008 messages for instances with un-initialized input pins.
- Fixed an issue in the VCD analysis where the dynamic results were inconsistent between runs when multiple dynamic analysis were performed using VCD, even with the same start and end times.
- Resolved an issue in which internal power table hooked up to internal power pin was not handled correctly when LIB2AVM_CLK_MSTATE option was turned on in standalone lib2avm run.
- Fixed a reporting issue in `adsRpt/<design_name>.power.rpt`, where the library name (column #16) was incorrect when GSR points to multiple libraries (different corners).

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Extraction

- Fixed an issue where the etch width calculated for the internal wire segments of a sliced metal layer/layers defined with the GSR keyword, INFO_LAYERS, were incorrect.
- Fixed an issue in which RedHawk failed to handle TSV models with grounded Rsub and reported incorrect extraction results.

Simulation

- Fixed an issue where incorrect results were shown in power rampup movie GUI for a single domain design.
- Fixed an issue in which RedHawk calculated incorrect input slew for a macro. Difference in switching power value can be expected due to this.

Power EM

- Fixed an issue in which RedHawk reported incorrect EM check values due to a bug in determining the combined wire widths for parallel slivers on different but coplanar layers.
- Fixed an issue in which Via-Via spacing failed to ignore some overlapping MFILL polygons.

Signal EM

- Fixed an issue in Signal EM, where the current given through EM_CUSTOM_CURRENT_FILE was not honored for primary input nets.
- Fixed an issue where the pin capacitance of an inout pin was not assigned properly even though the capacitance value was present in the .lib file.
- Fixed an issue in which incorrect slot via total area was reported for cases with very large via cuts.
- Fixed an issue where certain non-master pseudo vias with no EM value were being assigned a FIT value of 1 instead of 0.

ESD

- Fixed a bug in which PathFinder-S reported incorrect currents on the 'from'/'to' node .
- Fixed an issue in which PathFinder-S failed to create pin pairs non-aligned ESD pins.
- Fixed an issue in clamp node selection, where both POINT and BOX were specified for clamp locIDs in clamp file, PathFinder executed on POINT, and it the chosen POINT was outside the BOX.

GUI

- Fixed an issue in which non-linear color patterns were displayed in the EM map on large metals geometries with slots or gaps
- Fixed an issue in displaying the Timing Window of sequential cells and memory cells in RH GUI. Earlier output pin timing window was displayed instead of Clock pin timing window for such cells.

GDS2RH

- Fixed a bug in which the wildcard character (*) was not supported properly in GDS2RH keyword "GRAY_BOX_CELLS".

DMP

- Fixed an issue in which false TSV shorts were reported during Interposer extraction in DMP flow.

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- Fixed an issue where RedHawk 3DIC flow failed to handle the coordinates in BLOCK_POWER_ASSIGNMENT declared through BLOCK_POWER_MASTER_CELL GSR keyword in die's gsr file.
- Fixed the following issue in `perform rescalc -loopmode` reporting in DMP flow:
    - Individual records for each pin for certain instances in the report
    - The report format not following the table header when only one instance was provided in the query list.
- Fixed an issue in which RedHawk DMP failed to report all instances in the final report when 'perform res_calc -instFile <inputFile> -o <output> -pin vdd_ext' was repeated as shown in below example:

```
'perform res_calc -instFile sec_inst_list_vdd1.txt -o sec_pg.rpt -pin
vdd1'

'perform res_calc -instFile sec_inst_list_vdd2.txt -o sec_pg.rpt -pin
vdd2'
```

CPA

- Fixed polygon merge issue, user can set ignore small area in Polygon Merge tab now.
- Fixed pad current file with no space after comma and caused co-simulation failed.

CTA

- Fixed an issue where user was unable to create prototype pcb.
- Fixed MAPDL path setup issue, user does not need to set MAPDL path for each run.

Selfheat

- Fixed missing layer thickness information from CTMheader.

Other

- Fixed an issue in which RedHawk Explorer was reporting a GEN-025 error because of a missing apache.rc0Net file in case of perfect SPEF annotation.
- Fixed an issue in which the current waveforms were broken when VG mode APL was imported in LG mode using aqua.
- Fixed an issue in which APL failed to generate Current and Cdev profiles with a .lib file which had no functions or inputs. The fix is enabled with '`ENHANCED_LIB_SUPPORT` 1' keyword in apldi.config.
- Fixed a bug in CDEV creation when pwell was biased to a positive value.
- **apldo.** Fixed an issue in which "`plot voltage -ldo`" command was incorrectly reporting only the first pin waveform instead of both input and output waveforms..
- Fixed an issue in which RedHawk-3DIC failed to honor the cell names in the custom libs for early analysis in pre-processing BPA cells.
- Fixed the inconsistency between apache.apldi and final APL data samples when configuration keyword `SAMPLE_RANGE_LIB 1` was set to 1.
- Resolved an issue in which scan pins were not stored in the 'vector' information of APL current files when SCANMODE was turned on for flip-flop cells with scan logic

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